

RiskLens

AI-Powered Credit Risk Intelligence Platform

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Streamlit · LightGBM · SHAP · Groq LLaMA 3.3

Use Case & Business Value

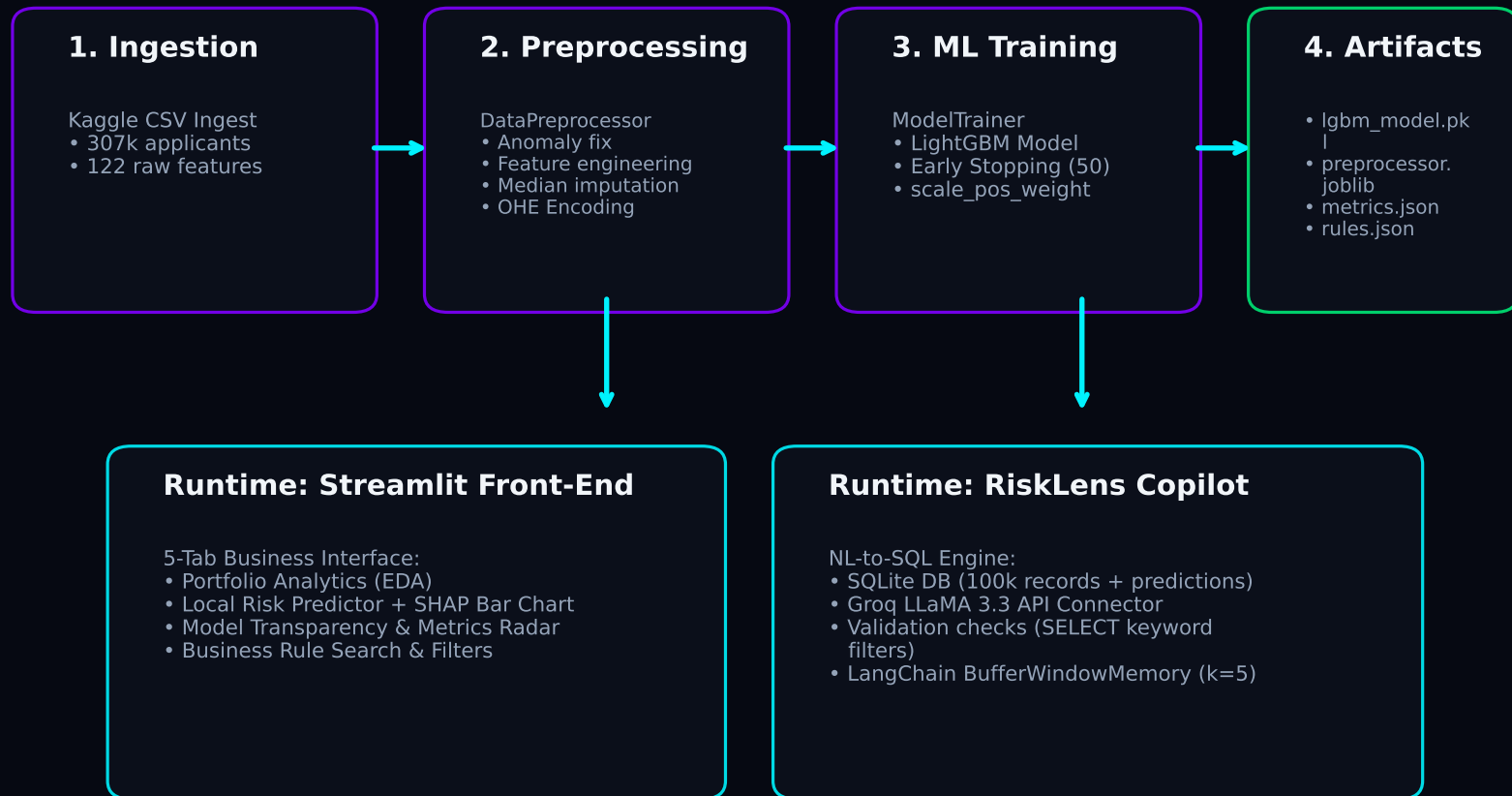
The Credit Risk Challenge

- High Default Exposure:
Financial institutions face heavy losses from unpaid loans.
- Opposing Objectives:
Underwriters need accurate predictions, but legacy models lack granular resolution.
- Black-Box Compliance Issues:
Modern ML (e.g., LightGBM) yields high performance but remains unexplainable, failing regulatory audits (like SR 11-7).
- Data Inaccessibility:
Risk managers struggle to explore portfolio metrics without writing complex SQL queries.

The RiskLens Core Solution

- ✓ Calibrated Risk Predictor:
Real-time default probability scoring using optimized thresholds.
- ✓ Regulatory Auditability via Rules:
Post-hoc surrogate decision tree maps complex predictions into human-readable if/then decision rules.
- ✓ Explainability at Scale:
Global & local feature attributions using SHAP explainers.
- ✓ RiskLens Copilot (NL-to-SQL):
Allows business teams to query SQLite portfolio records in natural language without technical knowledge.

Platform System Architecture



Model Performance & Imbalance Strategy

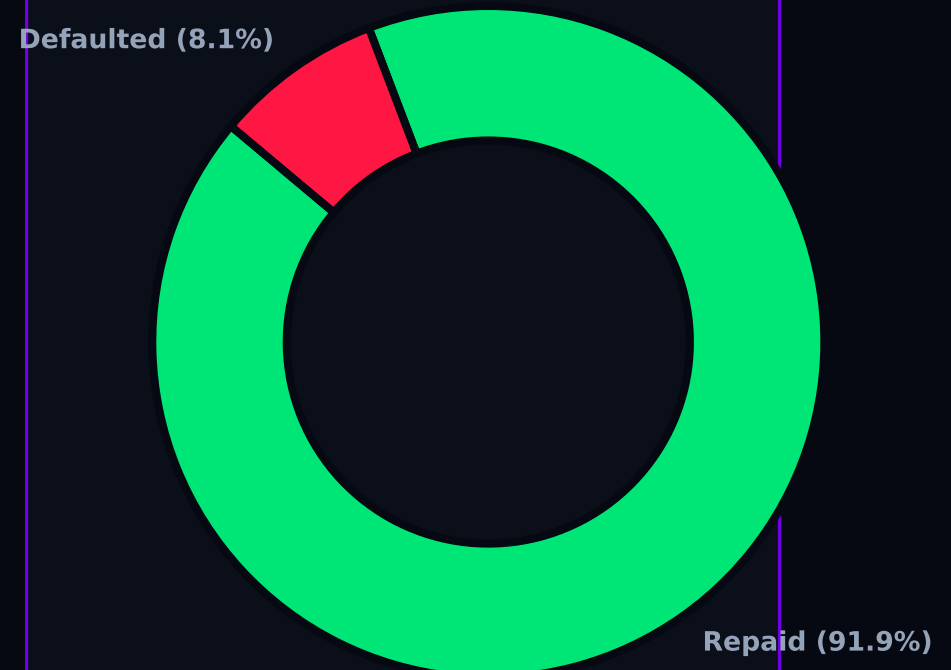
Model Metrics Report

- ROC-AUC Score: 0.7683
- PR-AUC Score: 0.2591 (3.2x over random baseline)
- F1 Score: 0.3202
- Calibrated Threshold: 0.6815
- Overall Validation Accuracy: 86.0%

Class Imbalance Strategy

- The Challenge:
Severe skewness (~92% repaid, ~8% defaulted).
A naive model achieves 92% accuracy by approving everyone.
- scale_pos_weight Penalty:
Loss function penalizes missed defaults ~11.3x more heavily.
- F1-Optimal Thresholding:
Precision-recall curve search anchors classification at score ≥ 0.6815 to maximize minority-class F1.

Applicant Default Imbalance (307k)



Model Transparency & Rules Engine

SHAP Local & Global Explanations

- True Transparency:
RiskLens uses shap.TreeExplainer to compute exact Shapley attribution values for applicant features.
- Local Risk Shift:
Attributions show how each feature pushes individual probability higher (red, increases risk) or lower (green, reduces risk) relative to the dataset expected base probability.
- Top Global Features:
 1. External Credit Score 2 (EXT_SOURCE_2)
 2. External Credit Score 3 (EXT_SOURCE_3)
 3. Avg External Score (EXT_SOURCE_MEAN)
 4. Age (AGE_YEARS)
 5. Leverage Ratio (CREDIT_INCOME_RATIO)

Surrogate Rule Derivation

- Method:
Trains a shallow DecisionTree (max depth=4) on LightGBM binarized targets to model the complex model's boundary.
- Key Metrics:
 - Extracted Rules: 16 human-readable paths
 - Surrogate Fidelity: 85.73% agreement rate
- Calibrated Band Rules:
Assigns leaves to Low, Medium, and High bands based on average leaf default probability (optimal thresholds).
- Sample Rule Output (Medium Risk):
IF External Credit Score 2 $\leq$ 0.4305
AND External Credit Score 3 $>$ 0.31
AND Employment Duration $>$ 4.5 years
--> RISK BAND: Medium (Confidence: 58.9%, Support: 4.4%)

RiskLens Copilot — Conversational Analyst

Copilot Agent Core Capabilities

- Zero-Code Data Access:
Translates natural language questions into executable SQL queries, runs them against SQLite, and summarizes output.
- Multi-Turn Memory Context:
Utilizes LangChain ConversationBufferWindowMemory (k=5) to retain query sequence for subsequent follow-up tasks.
- Schema-Anchored Safeguards:
Injects DDL and standard category values inside prompts to block column hallucination.
- Two-Layer Injection Defenses:
 - Validates query structure (forces SELECT statements only).
 - Blocks destructive commands (DROP, DELETE, UPDATE, ALTER, etc.).
 - Caps output fetching to 100 records to prevent memory overflow.

Verified Query Patterns & Examples

- ✓ Query Pattern 1: Aggregates by Class
 - * 'What is the average income of applicants who defaulted?'
 - * SQL: `SELECT AVG(AMT_INCOME_TOTAL) FROM applications WHERE TARGET=1`
- ✓ Query Pattern 2: Categorical Default Rates
 - * 'Show default rate by education type'
 - * SQL: `SELECT NAME_EDUCATION_TYPE, AVG(TARGET)*100 FROM applications...`
- ✓ Query Pattern 3: Filter & Count Queries
 - * 'How many female applicants have more than 2 children?'
 - * SQL: `SELECT COUNT(*) FROM applications WHERE CODE_GENDER='F'...`
- ✓ Query Pattern 4: Top-N Leaders
 - * 'What are the top 10 highest credit amounts?'
 - * SQL: `SELECT AMT_CREDIT FROM applications ORDER BY AMT_CREDIT DESC...`
- ✓ Query Pattern 5: Multi-band comparisons
 - * 'Show average external credit score by risk band'
 - * SQL: `SELECT RISK_BAND, AVG(EXT_SOURCE_MEAN) FROM applications...`

